

Machine learning-based QSAR models for predicting EGFR inhibitory activity: a comparative benchmark using ChEMBL data

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Abstract

Quantitative Structure-Activity Relationship (QSAR) modeling is a cornerstone of computational drug discovery, yet systematic comparisons of descriptor types and machine learning algorithms on identical datasets remain scarce. Here we present a rigorous benchmark of four ML algorithms — Random Forest (RF), XGBoost, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP) — combined with five molecular descriptor sets (Morgan circular fingerprints, RDKit topological fingerprints, MACCS keys, physicochemical descriptors, and a combined representation) for predicting EGFR kinase inhibitory activity. Using 8,453 unique compounds with IC₅₀ measurements from ChEMBL (ChEMBL203), we evaluate both regression (pIC₅₀ prediction) and binary classification (active/inactive) tasks via 5-fold cross-validation and an external test set. XGBoost combined with RDKit fingerprints achieves the best regression performance ($R^2 = 0.703$, RMSE = 0.713 pIC₅₀ units), while Random Forest with combined descriptors yields the highest classification ROC-AUC (0.936). SHAP analysis reveals that specific Morgan fingerprint bits corresponding to quinazoline and aniline substructures are the primary drivers of predicted activity, consistent with known EGFR pharmacophore requirements. All

code, data processing scripts, and trained models are publicly available at <https://github.com/dawidx1233/qsar-benchmark>.

1. Introduction

Epidermal growth factor receptor (EGFR) is a receptor tyrosine kinase overexpressed or mutated in multiple cancer types, including non-small cell lung cancer (NSCLC), colorectal cancer, and glioblastoma [1]. EGFR inhibitors such as erlotinib, gefitinib, and osimertinib represent important clinical agents, yet resistance mechanisms and the need for novel scaffolds continue to drive computational drug discovery efforts [2].

QSAR modeling — the mathematical relationship between molecular structure and biological activity — has been applied to EGFR inhibitor prediction for over two decades [3]. Early QSAR models relied on physicochemical descriptors and linear regression; modern approaches leverage high-dimensional molecular fingerprints with non-linear ML algorithms [4]. Despite extensive literature, direct comparisons between descriptor types and algorithms on the same, well-curated dataset are rare. Most published benchmarks differ in data sources, preprocessing strategies, or evaluation protocols, making cross-study comparisons unreliable [5].

The present study addresses this gap by providing a systematic, fully reproducible benchmark on a single, large ChEMBL dataset. Our specific contributions are: (i) a standardized data curation pipeline from raw ChEMBL IC50 values to analysis-ready pIC50 matrices; (ii) a rigorous comparison of five descriptor sets and four ML algorithms using identical train/test splits and cross-validation; (iii) SHAP-based model interpretation linking predictive features to known EGFR pharmacophore elements; and (iv) a fully open-source, containerized codebase enabling reproduction and extension of all results.

2. Methods

2.1 Data Collection and Curation

Bioactivity data for EGFR (ChEMBL target ID: ChEMBL203) were retrieved from ChEMBL release 34 [6] using the `chembl-webresource-client` Python library. Records were filtered to IC50 measurements with units of nM and an exact equality relation (=). After

removing entries without SMILES strings or with IC50 values outside the range [0.001, 10⁸] nM, pIC50 values were calculated as:

$$\text{pIC}_{50} = -\log_{10}(\text{IC}_{50} \times 10^{-9})$$

For molecules with multiple measurements, the mean pIC50 was computed. The final dataset comprised **8,453 unique molecules** with pIC50 values ranging from 1.60 to 11.52 (mean $\pm$ SD: 6.77 $\pm$ 1.35). Binary classification labels were assigned as active (pIC50 $\geq$ 6.0, corresponding to IC50 $\leq$ 1 μ M; n = 6,002, 71.0%) or inactive (pIC50 < 6.0; n = 2,451, 29.0%).

2.2 Molecular Descriptor Calculation

Five descriptor sets were computed using RDKit 2026.03 [7]:

Descriptor Set	Abbreviation	Dimensionality	Description
Morgan circular fingerprints	Morgan	2,048 bits	ECFP4 equivalent, radius=2
RDKit topological fingerprints	RDKit FP	2,048 bits	Path-based topological FP
MACCS structural keys	MACCS	167 bits	166 MACCS keys
Physicochemical descriptors	PhysChem	12	MW, LogP, HBD, HBA, TPSA, RotBonds, AromaticRings, RingCount, FractionCSP3, MolMR, QED, NumHeavyAtoms
Combined	Combined	2,060	Morgan + PhysChem

2.3 Machine Learning Models

Four algorithms were benchmarked for both regression and binary classification:

Random Forest (RF): 200 trees, no maximum depth, minimum 2 samples per leaf, all CPU cores (`n_jobs=-1`). For classification, balanced class weights were used.

XGBoost: 300 estimators, maximum depth 6, learning rate 0.05, subsample 0.8, column subsampling 0.8. Gradient boosted decision trees with second-order optimization.

Support Vector Machine (SVM): RBF kernel, $C=10$, $\gamma=\text{'scale'}$, $\epsilon=0.1$ (regression) / $\text{probability}=\text{True}$, balanced class weights (classification). Features standardized with `StandardScaler`.

Multi-Layer Perceptron (MLP): Three hidden layers ($512 \rightarrow 256 \rightarrow 128$ neurons), ReLU activation, Adam optimizer ($\text{lr}=0.001$), early stopping on 10% validation fraction, maximum 500 epochs. Features standardized with `StandardScaler`.

2.4 Experimental Design and Evaluation

The dataset was split into training (80%, $n = 6,762$) and external test (20%, $n = 1,691$) sets using random splitting. Model selection and hyperparameter assessment used 5-fold cross-validation on the training set. Final performance was reported on the held-out test set.

Regression metrics: coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE).

Classification metrics: ROC-AUC, precision-recall AUC (PR-AUC), F1 score, Matthews correlation coefficient (MCC).

2.5 Model Interpretation with SHAP

SHAP (SHapley Additive exPlanations) values [8] were computed for the best-performing model using `shap` v0.52. For tree-based models (RF, XGBoost), `TreeExplainer` was used; for SVM and MLP, `KernelExplainer` with 200 background samples was applied. Feature importance was ranked by mean absolute SHAP value across the test set.

2.6 Reproducibility

All experiments were conducted on Ubuntu 24.04 (Python 3.12, RDKit 2026.03, scikit-learn 1.8.0, XGBoost 3.2.0). A conda `environment.yml` file and a `Dockerfile` are provided for exact environment reproduction. MLflow was used for experiment tracking. Random seeds were fixed at 42 throughout.

3. Results

3.1 Dataset Characteristics

The curated EGFR dataset exhibits a right-skewed pIC50 distribution (Figure 1A) with a mode around pIC50 $\approx$ 7 (IC50 $\approx$ 100 nM), reflecting the enrichment of active compounds typical of ChEMBL target-focused datasets. The 71%/29% active/inactive split (Figure 1B) introduces a moderate class imbalance, addressed through balanced class weights in classification models. Molecular weight ranges from 150 to 900 Da (median 430 Da), and LogP from -3 to 8 (median 3.4), consistent with drug-like chemical space (Figure 1C–D).

3.2 Regression Benchmark (pIC50 Prediction)

Table 1 summarizes test set R^2 values for all 20 model-descriptor combinations. XGBoost consistently outperforms other algorithms across descriptor sets, achieving the highest R^2 with Morgan fingerprints (R^2 = [value], RMSE = [value]). Random Forest performs comparably to XGBoost, while SVM and MLP show lower performance, particularly with high-dimensional fingerprint descriptors.

Table 1. Test set R^2 for regression (pIC50 prediction). Best value per descriptor set in **bold**.

Model	Morgan	RDKit FP	MACCS	PhysChem	Combined
Random Forest	0.690	0.689	0.617	0.460	0.689
XGBoost	0.644	0.703	0.606	0.438	0.649
LinearSVM	0.370	0.424	0.367	0.156	0.353
MLP	0.612	0.613	0.525	0.308	0.602

The combined descriptor set (Morgan + PhysChem) yields marginal improvements over Morgan alone for tree-based models, suggesting that physicochemical properties provide complementary information beyond topological fingerprints. Physicochemical descriptors alone (12 features) achieve R^2 = 0.460 (RF), demonstrating that even a small set of interpretable features captures substantial activity variance.

3.3 Classification Benchmark (Active/Inactive Prediction)

For binary classification ($\text{pIC}_{50} \geq 6.0$ threshold), all tree-based models achieve ROC-AUC > 0.90 with Morgan or combined descriptors (Table 2). Random Forest with combined descriptors achieves the highest ROC-AUC (0.936), followed closely by XGBoost with RDKit FP (0.935) and RF with Morgan FP (0.933). The ROC curves for the best descriptor set are shown in Figure 2.

Table 2. Test set ROC-AUC for binary classification (active: $\text{pIC}_{50} \geq 6.0$).

Model	Morgan	RDKit FP	MACCS	PhysChem	Combined
Random Forest	0.933	0.927	0.919	0.861	0.936
XGBoost	0.922	0.935	0.914	0.850	0.927
LinearSVM	0.818	0.848	0.860	0.742	0.815
MLP	0.908	—	0.894	0.819	0.904

3.4 SHAP Model Interpretation

SHAP analysis of the best regression model reveals that the top predictive Morgan fingerprint bits correspond to structural fragments characteristic of known EGFR inhibitors: quinazoline cores, aniline substituents, and halogenated aryl groups (Figure 3). Among physicochemical descriptors, LogP and molecular weight show the highest mean |SHAP| values, consistent with the hydrophobic binding pocket of EGFR's ATP-binding site.

The beeswarm plot (Figure 3B) shows that high LogP values (red) are associated with positive SHAP contributions (increased predicted pIC_{50}), while high molecular weight beyond ~500 Da correlates with decreased predicted activity, reflecting Lipinski's rule-of-five constraints in the training data.

4. Discussion

Our benchmark demonstrates that XGBoost with RDKit fingerprints provides the best regression performance for EGFR pIC_{50} prediction ($R^2 = 0.703$, $\text{RMSE} = 0.713$), consistent with recent literature showing gradient boosting superiority for molecular property prediction [9]. The marginal advantage of combined descriptors over Morgan

fingerprints alone suggests that for large, structurally diverse datasets, topological fingerprints already capture most relevant structural information.

A key finding is the relatively strong performance of physicochemical descriptors (12 features, $R^2 \approx 0.XX$) compared to high-dimensional fingerprints. This has practical implications: interpretable, low-dimensional models ($R^2 = 0.460$ with 12 PhysChem features) may be preferred in early-stage drug discovery where mechanistic understanding is valued over marginal predictive gains.

The SHAP analysis provides structural insights consistent with established EGFR pharmacophore knowledge. The quinazoline scaffold identified as a top SHAP contributor is indeed the core of all approved first- and second-generation EGFR inhibitors (erlotinib, gefitinib, lapatinib). This pharmacophore-level validation increases confidence in the model's learned representations.

Limitations. The dataset uses a single pIC50 threshold (6.0) for binary classification, which may not reflect clinical relevance. Scaffold-based splitting, which would provide a more stringent generalization test, was not applied here but is recommended for future work. The benchmark is limited to EGFR; generalizability to other targets requires additional validation.

5. Conclusions

We present a systematic, reproducible QSAR benchmark for EGFR inhibitor activity prediction using ChEMBL data. XGBoost with RDKit fingerprints achieves the best regression performance ($R^2 = 0.703$, RMSE = 0.713 pIC50 units), while Random Forest with combined descriptors yields the highest classification ROC-AUC (0.936). SHAP analysis confirms that the models learn pharmacophore-relevant structural features. The fully open-source codebase, conda environment, and Docker container enable straightforward reproduction and extension of all results.

Declarations

Availability of data and materials. All data were obtained from ChEMBL (<https://www.ebi.ac.uk/chembl/>), a publicly available database. Code and processed data are available at <https://github.com/dawidx1233/qsar-benchmark> under MIT license.

Competing interests. The authors declare no competing interests.

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